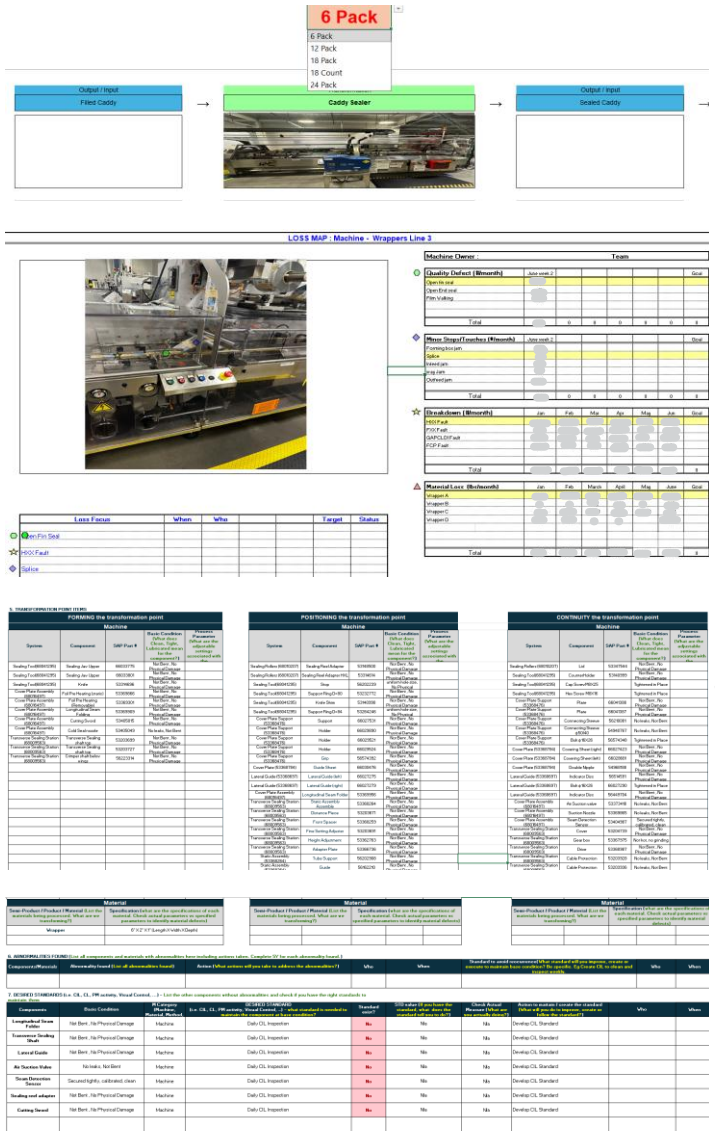
MANUFACTURING ENGINEERING INTERNSHIP | MONDELĒZ INTERNATIONAL | CLIF BAR – TWIN FALLS, IDAHO - PRANAV NADIG

PROCESS MAPPING UNDER 10+5

About Me & This Internship:
During my process engineering internship, I supported operations through process improvement and quality-focused initiatives.
My primary project centered on process mapping through the 10+5 Quality Management framework. I conducted a detailed analysis of the wrapper's equipment, breaking the process into individual steps to identify opportunities for standardization, improvement and quality.
This project strengthened my understanding of structured process analysis and its role in driving operational excellences and continuous improvement

Project Execution:
My project evolved through several stages as I applied process mapping principles to improve documentation and support continuous improvement.

- **Enhanced Existing Process Maps**
  - Expanded the existing Line 3 process map by adding the manufacturing portion and documenting Critical Process Parameters (CPPs) where applicable.
  - Included manufacturing information for every Line 3 product.
  - Developed a drop-down interface that allows users to view process information for individual products quickly. Although not a direct requirement of the 10+5 methodology, this feature improved usability for operators and received positive feedback from the manufacturing team.
- **Adopted the New Process Mapping Standard**
  - Following training with the Corporate Quality standard owner, transitioned to the updated process mapping standard and incorporated the new methodology into future work.
- **Developed a Wrapper Loss Map**
  - Created a loss map for the wrapper machines using available production data.
  - Collaborated with the Line 3 Process Engineer to identify missing information and support the implementation of a shop floor data collection form, enabling more complete loss analysis.
- **Documented Wrapper Working Principles**
  - Developed detailed working principles for the wrapper machines, with an emphasis on understanding and reducing open seal defects. Identified active areas for improvement across the region.
  - Classified each improvement according to its corresponding transformation category to support standardized knowledge sharing and best practices.



Acknowledgements: Crystal Pereira Rodriguez (TIM Sr. Engineer), Riley Monson (Process Engineer), Isaac Stokes (Process Engineer), Matt Curnow (Line Manager), Michael Kartrick (Corporate Quality), Lucas Conway (Parts Room Manager), Nate Swainston (Project Engineer), Mitch Eckles (Transactional Data Coordinator), Michael Wilson (Process Engineer), Christopher Park (Continuous Improvement Engineer), Ludy Angulo (TIM Manager), David Solomon (Continuous Improvement Engineer)

Conclusion and Future Work:
This project established the foundation for implementing standardized process mapping under the 10+5 Leadership Principles. By enhancing process documentation, developing wrapper loss maps, and creating equipment working principles, the project provided a structured approach to understanding manufacturing processes and identifying opportunities for continuous improvement.
As the project transitions to the next person, future work will include:

- Expanding loss mapping to the next major production losses.
- Developing working principles for additional equipment.
- Identifying reliable sources for process and equipment data.
- Creating standardized methods for collecting information that is not readily available on the production floor.
- Continuing to refine and standardize process mapping documentation to support long-term operational excellence.

Additional Contributions & Professional Development: Throughout my internship, I had the opportunity to contribute to several initiatives beyond my primary project, allowing me to broaden my technical knowledge and gain a deeper understanding of manufacturing operations.
Project Engineering Support: Assisted a Project Engineer in developing the initial framework for a speed matrix to track the operating speeds of individual pieces of equipment across the production line.
10+5 Leadership Implementation: Contributed to the development of 10+5 training cycles and participated in both Cycle 1 and Cycle 2, helping support the rollout of the program.
Cross-Functional Exposure: Attended cross-functional meetings with my manager that extended beyond the scope of my internship, providing insight into the different operational pillars and how they work together to run a manufacturing facility.
Professional Networking: Connected with engineers, managers, and leaders across the plant to learn from their experiences, gain industry perspectives, and receive valuable career guidance.

These experiences allowed me to develop not only my technical skills but also a broader understanding of manufacturing operations, continuous improvement, and cross-functional collaboration.